G11 PHYSICS

Practice Problem Set

Vectors

1. Palden skis 1.00 km north and then 2.00 km east on a horizontal snow field. How far and in what direction is she from the starting point?

2. Two force vectors, $\vec{a}$ and $\vec{b}$, have magnitudes $a = 2$ N and $b = 5$ N respectively. Which of the following could be the magnitudes of the difference vectors $\vec{a} - \vec{b}$? (There may be more than one correct answer.)

   a. 7 N

   b. 10 N

   c. -3 N

   d. 3 N

3. Let

$$\vec{a} = \frac{1}{2}\hat{x} - 3\hat{y} + 9\hat{z} \qquad \text{and} \qquad \vec{b} = -4\hat{x} + 3\hat{z}.$$

   (i) Find the angle between $\vec{a}$ and $\vec{b}$.

   (ii) Find the distance between $\vec{a}$ and $\vec{b}$.

4. Find the resultant of the four vectors shown in Fig. 1.

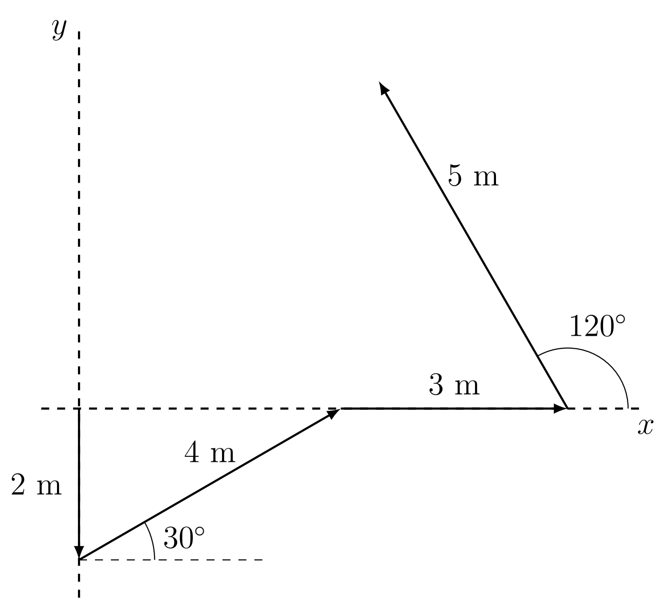


Fig. 1

Dimensional Analysis

1. It is observed that water waves on a very deep ocean travel with a speed v that depends upon their wavelength λ and the density ρ of the water, but not upon their amplitude. Use dimensional analysis to find how v depends upon λ and ρ.

2. The heat produced in a wire carrying an electric current depends on the current I, the resistance R, and the time t. Using dimensional analysis, derive an expression for the heat H produced. The dimensional formula of resistance is $[R] = ML^2I^{-2}T^{-3}$, and heat is a form of energy.

3. The acceleration due to gravity on the Moon is 1.68 $\mathrm{m/s^2}$. Convert this value to $\mathrm{cm/min^2}$.

4. If all the terms in an equation have the same units, is it necessary that they have the same dimensions? If all the terms in an equation have the same dimensions, is it necessary that they have the same units?

5. The special theory of relativity reveals that mass can be converted into energy. The energy E so obtained is proportional to certain powers of mass m and the speed of light c. Using dimensional analysis, derive a relation between E, m, and c.

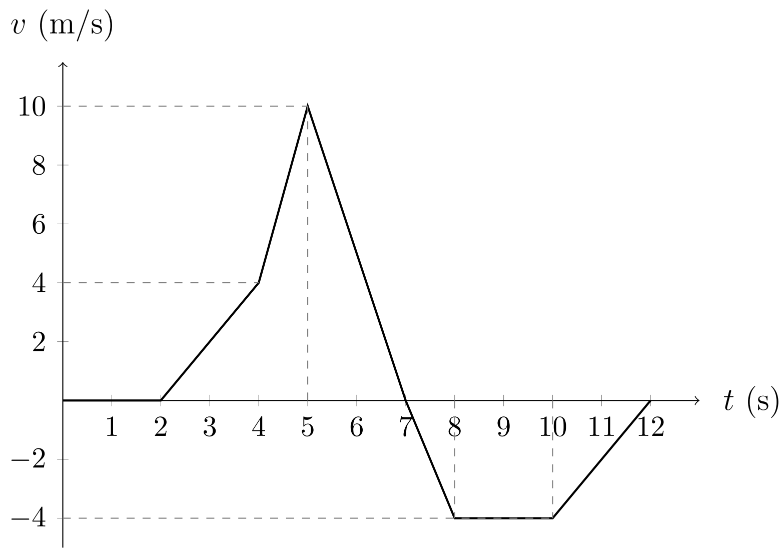
6. Test if the following equations are dimensionally correct:

   a. $f = \frac{1}{2\pi}\sqrt{\frac{mgl}{I}}$ b. $h = \frac{2S\cos\theta}{\rho r g}$

   where h = height (m), S = surface tension (N/m), ρ = density ($\mathrm{kg/m^3}$), r = radius (m), g = acceleration due to gravity ($\mathrm{m/s^2}$), l = length (m), m = mass (kg), f = frequency ($\mathrm{s^{-1}}$), and I = moment of inertia ($\mathrm{kg\,m^2}$).

Motion

1. Studying the following velocity–time graph, determine how far away the body is at $t = 10$ s.



2. Under what conditions is average velocity equal to instantaneous velocity?

3. A tiger is crouched 30 m to the east of a photographer's vehicle. At time $t = 0$ the tiger charges a deer and begins to run along a straight line. During the first 2.0 s of the attack, the tiger's coordinate x varies with time according to the equation

$$x(t) = 30\text{ m} + (5.0\text{ m/s}^2)\,t^2.$$

   a. Find the displacement of the tiger between $t_1 = 1.0$ s and $t_2 = 2.0$ s.

   b. Find the average velocity during the same time interval.

   c. Find the instantaneous velocity at time $t_1 = 1.0$ s by taking $\Delta t = 0.1$ s, then $\Delta t = 0.01$ s.

   d. Derive a general expression for the instantaneous velocity as a function of time, and from it find v at $t = 1.0$ s and $t = 2.0$ s.